AKASH RAO

PhD Student in Computer Science | Manifold Learning, Representation Geometry, Few-Shot and Zero-Shot Learning

Auburn, AL | 470.621.2259 | azr0187@auburn.edu | LinkedIn | GitHub | Google Scholar

EDUCATION

Auburn University, Auburn, AL *January 2025 – Expected Spring 2028*

PhD in Computer Science, advised by Dr. Sathyanarayanan N. Aakur (ORCA Lab)

Auburn University, Auburn, AL *August 2023 – August 2025*

MS Data Science and Engineering

Visvesvaraya Technological University, India *August 2017 – August 2021*

BE Computer Science and Engineering

PUBLICATIONS

1. **"FSP-DETR: Few-shot Prototypical Parasitic Ova Detection,"** Shubham Trehan, Udhav Ramachandran, Akash Rao, Ruth Scimeca, and S.N. Aakur *(Co-First Author, Published, IEEE/CVF WACV 2026)*
openaccess.thecvf.com (WACV 2026 paper)

Diagnosing parasitic infections from microscope images is hard when only a handful of labeled examples exist per parasite. Built FSP-DETR, a single object detection model that handles few-shot detection of known parasites, open-set recognition of unfamiliar ones, and zero-shot generalization to entirely new parasite types from just a few examples and curated a publicly released benchmark of 3,001 annotated images across 20 parasite species.

- Paired a class-agnostic DETR detection backbone with prototype-based classification to improve few-shot detection accuracy over the best existing method with only 5 labeled examples per parasite (mAP 0.066 vs. 0.039), outperforming eight competing object detection systems including YOLO and DETR.
- Added a background prototype rejection mechanism for open-set recognition, correctly flagging 3 out of 4 parasite species never seen during training (74% open-set recall) with zero labeled examples of them.
- Trained the embedding space with an energy-based objective (prototype matching + KL-divergence distillation + contrastive InfoNCE-style alignment) that generalized to two unrelated few-shot medical imaging tasks, ranking first on both blood cell and malaria detection using just 5 examples per class.

2. **"HyperTaxa: Multiscale Genome Representations for Generalized Microbial Taxonomic Inference,"** Akash Rao, Sushim Gupta, Vishalini Ramnath, Akhilesh Ramachandran, and S.N. Aakur *(First Author, Accepted, IEEE BIBM 2026)*
https://arxiv.org/abs/2610.25673

As DNA sequencing produces a flood of new organisms, identifying them becomes hard when a species is rare, poorly studied, or missing from reference databases. HyperTaxa tackles this by learning genome representations that can place a bacterial species into the tree of life even with no training examples, supporting zero-shot (ZSL), generalized zero-shot (GZSL), and hierarchical inference.

- Encoded genomes as multiscale hyperspectral k-mer transition fields aligned to a taxonomy-aware embedding, reaching 53%, 66%, and 75% zero-shot accuracy at genus, family, and order levels on unseen bacterial species (97-species TEPI benchmark), substantially above prior pseudo-image and transformer-graph baselines.
- Designed a two-stage Node2Vec-to-MDS embedding alignment that preserves genus-family-order hierarchy, reaching 64–70% generalized zero-shot (GZSL) accuracy at higher taxonomic ranks on a 1,110-species training / 219 unseen-species evaluation.
- Added cascaded genus-to-species inference (about 5% higher species accuracy) and showed via ablations that a lightweight CNN beats ResNet-50 and ViT-B/16, so embedding geometry matters more than model capacity.

3. **"Class Geometry as Supervision for Sample-Efficient Open-World Detection,"** Akash Rao, Zhou Chen, Revanth Reddy Palem, Udhav Ramachandran, Ruth Scimeca, and S.N. Aakur *(First Author, Under Review, AAAI 2027)*
https://arxiv.org/abs/2608.12698

Open-world detectors must recognize known objects, reject unfamiliar ones, and absorb new categories over time, but prototype-based detectors learn each class as an isolated anchor and ignore how classes relate. Class-Geometry Supervision (CGS) treats those relationships as a training signal, constraining the learned feature space to preserve real visual and semantic dissimilarities between classes for sample-efficient open-world, few-shot, and open-set detection.

- Applied CGS to novel-class insertion to place new categories into an existing detector by their visual/semantic relationships rather than as isolated anchors, improving novel-class detection accuracy roughly 2.5x (mAP 0.063 to 0.156) and recall from 0.26 to 0.37.

- Used the geometry-organized prototype space to sharpen open-set rejection of unknown objects, raising open-set accuracy 36% (mAP 0.045 to 0.061), and improved few-shot biomedical detection (5-shot mAP 0.066 to 0.076).
- Transferred CGS to the standard COCO open-world benchmark (OW-DETR) under frozen-head and full fine-tuning setups, increasing unknown-object recall across multiple task stages.

WORK EXPERIENCE

Graduate Teaching Assistant, Auburn University *Fall 2026 – Present*

- Teaching Assistant for CPSC 3700 (Software Engineering I) and CPSC 4970 (Special Topics in AI and Machine Learning).

Graduate Research Assistant, ICAMS *October 2024 – Fall 2026*

- Solely built an offline, private camera surveillance and streaming system (React, MQTT, Docker, Flask, Raspberry Pi) with motion detection and real-time video prioritization, deployed across two lab floors with no cloud dependency.
- Developed a local-first machine-telemetry platform (Next.js, FastAPI, PostgreSQL, MQTT, Docker) that routes live sensor data from CNC and manufacturing machines into session-based datasets, with a Redis-backed real-time dashboard.
- Led development of an MRP (Material Requirements Planning) platform (Next.js, FastAPI, PostgreSQL, Redis) for manufacturing workflows, inventory, and scheduling, with JWT role-based access control and bulk CSV/XLSX import pipelines.
- Built a lab job management application (React Native, Flask, PostgreSQL) integrated with the Paperless Parts API, enabling supervisors to create and assign incoming job orders while students view, track, and complete them.
- Served as technical lead and mentor for a team of undergraduate developers as the sole PhD student, overseeing code quality, architecture, and delivery across lab software projects.

Software Engineer, Dcode Solutions L.L.C. *October 2021 – June 2023*

- Built an end-to-end asset tracking system for the U.A.E. Ministry of Monitor and Control (Android/Java, MSSQL, React) with Honeywell barcode scanning for warehouse staff and a three-tier approval and audit workflow through a web dashboard.
- Engineered the system to operate fully offline for warehouses without internet, syncing to the server post-operation, and added GPS tracking for outdoor assets and RFID support alongside the primary barcode workflow.
- Served as solo technical lead integrating a Bartender and Excel based stocktaking solution into Dubai Duty-Free's systems and customized stocktake applications for major U.A.E. retailers (Sharaf DG, Koi, and others), including client training.

Software Engineer Intern, Scientia Innovations *October 2020 – December 2020*

- Independently built a cross-platform Flutter marketplace app for online courses, including third-party API integrations.

PROJECTS

Auburn-Regions Data Hackathon (Sponsored by Regions Bank)

- Built a credit risk model in Python (scikit-learn, pandas) predicting each loan applicant's probability of default from simulated banking data (30,000 training, 7,500 evaluation applicants), using relational features from linked credit account and prior lending history, reaching 0.77 validation ROC AUC (0.75 5-fold CV AUC, 0.095 Brier score).
- Translated predicted default probabilities into a three-tier lending policy (auto-approve, manual review, decline) that approved 56% of validation applicants at a 5.2% default rate versus 12.6% overall, turning a simulated portfolio loss into a profit.
- Delivered a fully reproducible, leakage-safe pipeline with fixed random seeds, SHA-256 data integrity checks, and SHAP model explanations, plus a responsible AI fairness audit across demographic groups kept out of the model.

Embedding Geometry of Vision-Language Models *Ongoing*

- Studying the embedding spaces of CLIP and BioCLIP vision-language models in PyTorch for zero-shot species recognition from images.

Automated Image Collection with Robotic Arm

- Developed a Python control pipeline for a Dobot robotic arm that autonomously captures multi-angle images for vision datasets.

SKILLS AND CERTIFICATIONS

Programming Languages: Python, Java, JavaScript, SQL

ML / AI: PyTorch, TensorFlow, scikit-learn, pandas, NumPy, SHAP, CLIP, BioCLIP, DETR; local LLMs (Ollama, Llama 3.2)

Representation Learning: manifold learning, embedding alignment (MDS, Node2Vec), prototype, metric, and contrastive learning

Web, Mobile & Backend: React, Next.js, Node.js, FastAPI, Flask, REST APIs, React Native, Flutter, Native Android

Databases, Infrastructure & Hardware: PostgreSQL, MSSQL, Redis, Docker, MQTT, Raspberry Pi, Arduino, barcode/RFID systems

Certifications: Machine Learning (Stanford University, Coursera), Industry 4.0, Docker for Developers